Worksheet B

Geometry, Trigonometry and Vectors

Edexcel International GCSE Mathematics Grades 6–8

Name: _____ Date: _____ 45 minutes 45 marks

Calculator allowed. Show all your working. Diagrams are not drawn accurately.

1 **[6]**

In triangle ABC, $AB = 8.4$ cm, $BC = 11.2$ cm and angle $ABC = 64°$.

(a) Work out the length of AC. Give your answer correct to 3 significant figures. [4]

(b) Work out the area of triangle ABC. Give your answer correct to 3 significant figures. [2]

2 **[4]**

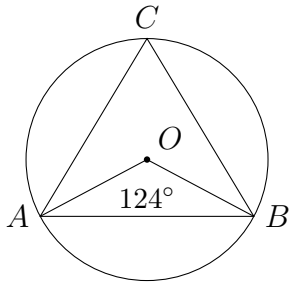
A cuboid has a rectangular base 8 cm by 6 cm and a height of 5 cm. The longest diagonal of the cuboid runs from one corner of the base to the opposite top corner.

(a) Work out the length of this diagonal. [2]

(b) Work out the angle between this diagonal and the base. [2]

© Bridge Up Tuition. Property of Bridge Up Tuition. Free for personal and classroom use; not to be sold or reproduced as your own.

3 **[4]**



A, B and C are points on a circle with centre O. Angle $AOB = 124°$.
(a) Find angle ACB. Give a reason. [2]

(b) Find angle OAB. Give a reason. [2]

4 **[4]**

Two vases are mathematically similar. The smaller vase is 12 cm tall and the larger is 18 cm tall.
(a) The surface area of the smaller vase is 80 cm^2. Work out the surface area of the larger vase. [2]

(b) The volume of the smaller vase is 240 cm^3. Work out the volume of the larger vase. [2]

5 **[5]**

OAB is a triangle. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. M is the midpoint of AB.
(a) Write $\overrightarrow{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [1]

(b) Show that $\overrightarrow{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$. [2]

P is the point on OM such that $OP : PM = 2 : 1$.
(c) Find $\overrightarrow{AP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. Give your answer in its simplest form. [2]

© Bridge Up Tuition. Property of Bridge Up Tuition. Free for personal and classroom use; not to be sold or reproduced as your own.

6 **[3]**

Solve $3\cos x = 1$ for $0° \leq x \leq 360°$. Give your answers correct to 1 decimal place.

7 **[4]**

A sector of a circle has radius 10 cm and angle 80°. A chord joins the two ends of the arc, cutting off a segment. Work out the area of the segment (the region between the chord and the arc). Give your answer correct to 3 significant figures.

8 **[3]**

Show that $\dfrac{5+\sqrt{2}}{3-\sqrt{2}}$ can be written in the form $a + b\sqrt{2}$ where a and b are fractions.

9 **[4]**

A circle has equation $x^2 + y^2 = 25$. The point $P(3, 4)$ lies on the circle. Find an equation of the tangent to the circle at P. Give your answer in the form $ax + by = c$ where a, b and c are integers.

© Bridge Up Tuition. Property of Bridge Up Tuition. Free for personal and classroom use; not to be sold or reproduced as your own.

10 **[6]**

A boat sails from A to B on a bearing of 065° for 12 km. It then sails from B to C on a bearing of 140° for 9 km.

(a) Work out the length AC. [4]

(b) Work out the bearing of C from A. [2]

11 **[2]**

Without a calculator, work out the exact value of $\cos 30° \times \tan 60°$.

© Bridge Up Tuition. Property of Bridge Up Tuition. Free for personal and classroom use; not to be sold or reproduced as your own.